MIL-C-48147 (MU)
18 April 1973
SUPERSEDING
PA-PD-2870
8 June 1966

MILITARY SPECIFICATION

CASTINGS, FERRITIC MALLEABLE IRON FOR M151 WARHEAD

1. SCOPE

1.1 This specification covers one grade of ferritic malleable iron castings for ammunition components.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification For.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1168 - Lot numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

FSC: 1340

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Melting Process.-The iron shall be made by one of the following processes: air-furnace, open hearth, electric-furnace or any of the multiplex processes.

3.2 Molding.-Molds and cores shall be made on semi-automatic or automatic molding equipment.

3.3 Heat Treatment.-Castings shall be annealed in a controlled atmosphere furnace. Heat treatment shall be performed on the whole casting in such a manner as to result in uniform mechanical properties throughout the individual part and throughout the heat treat lot.

3.3.1 After Heat Treatment.-After heat treatment, heating of the castings above the second stage annealing temperature shall not be permitted.

3.4 Mechanical Properties.-The castings shall meet the mechanical property requirements shown in Table I.

TABLE I

Yield Strength (psi) Minimum (0.2 percent offset)	Tensile Strength Minimum (psi)	Elongation Minimum (%)	Brinell* Hardness
28,000	48,000	10	110 to 156

*Brinell hardness range is advisory only.

3.5 Foundry Control.-Castings shall be produced under foundry control approved by the procuring activity.

3.6 Soundness.-The soundness of castings shall meet standards furnished or approved by the procuring activity using inspection methods specified in the item specification and by the procuring activity.

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3.7 Repair of Castings.-Defective castings shall not be plugged or welded.

3.8 Mottle.-Castings shall be free of mottle (primary graphite) when tested in accordance with 4.4.3.

3.9 Identification of Castings.-Castings representing a heat treat lot shall be identified in a manner approved by the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be

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required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions for Production.

4.2.1 Lot Formation.-The term "inspection lot" as used in this specification, is defined as an essentially homogeneous collection of units or products from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number.

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4.2.2 Heat-treatment Lot.-To determine compliance with the mechanical properties, 3.4, any quantity of castings not to exceed four (4) hours' production of the final heat-treatment in the same furnace, for the same length of time or at the same conveyor speed, shall be considered as a heat-treatment lot. If a batch type heat treat furnace is used, any quantity of 5,000 castings or less shall be considered a heat-treatment lot. The lot thus constituted shall be the basis for selection of sample castings to be used for tension tests, 4.3.1.2.

4.3 Testing

4.3.1 Mechanical Properties

4.3.1.1 Hardness.-In order to insure uniformity of heat-treatment and to establish a hardness range, to assure compliance with specified mechanical properties each casting after final heat-treatment shall be subjected to a hardness test with a Brinell Hardness Tester at a location on the casting specified or approved by the procuring agency.

4.3.1.1.1 Alternate hardness testing.-A direct reading Brinell Hardness Testing Machine may be substituted for the Brinelloscope method, provided the same diameter ball is used and the same load is applied as is used in Standard Brinell testing, and provided that at least 2000 castings are tested by both methods to establish a correlation between the two methods. After a correlation has been established, the diameter of the indentation in every tenth casting shall be measured with the Brinelloscope to assure continued correlation.

4.3.1.2 Tension Test.-Two castings from each lot representing the hardest and softest castings as determined by the hardness test outlined in 4.3.1.1 or 4.3.1.1.1 shall be subjected to the tension test. One specimen from each casting shall be prepared as specified in 4.4.1.2 and tested. If all specimens comply with the requirements of 3.4, the hardness range for the lot will be established. Failure of one or more tension test specimens to comply with the requirements shall be cause for rejection of the lot represented by the specimen.

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4.3.1.3 Retest.-When one or more tension test specimens fail to comply with the requirements of 3.4 and do not qualify for retest under 4.4.1.2.3, a new hardness range shall be established. The rejected lot may at the direction of the contractor, be reheat-treated prior to establishing a new hardness range. The tension test, in compliance with 4.3.1.2 shall be conducted to determine the acceptability of the new hardness range. When the hardness range has been established, all castings whose values fall outside this range shall be rejected.

4.3.2 Soundness.-Each casting shall be tested in accordance with 4.4.2 or as prescribed by the procuring activity.

4.3.3 Mottle.-A minimum of two castings selected for every thirty minutes or fraction thereof of continuous pouring time shall be tested in accordance with 4.4.3. Failure of any sample to comply with 3.8 shall be cause for rejection of all castings poured since the last acceptable sample was taken.

4.4 Test Methods and Procedures

4.4.1 Mechanical Properties

4.4.1.1 Hardness.-The Brinell hardness test shall be performed in accordance with procedures prescribed in Federal Test Method Standard No. 151. Due to the nature of the microstructure of ferritic malleable iron, indenter balls of less than ten millimeter diameter shall not be used.

4.4.1.2 Tension Test.-From each sample selected in accordance with 4.3.1.2, a tension test specimen shall be taken longitudinally from each sample, or as prescribed by the procuring activity.

4.4.1.2.1 Tension test specimen.-The specimen shall be the largest obtainable standard round specimen prescribed by Federal Test Method Standard No. 151 (type R1 thru R5) which the size of the casting and location of the specimen will permit. If the size or geometry of the casting does not permit a standard round specimen, a specimen as specified or approved by the procuring activity shall be used.

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4.4.1.2.2 Yield strength, tensile strength, and elongation.- Yield strength shall be determined by the offset method as defined in Federal Test Method Standard No. 151. The offset shall be 0.2 percent (0.002 inches per inch of gage length). Tensile strength and elongation on each specimen shall be determined by utilizing the criteria for these tests prescribed by Federal Test Method Standard No. 151.

4.4.1.2.3 Replacement test specimens.-If a tension test specimen fails to meet the requirements of 3.4 due to an identifiable defect in the tensile specimen which is acceptable under provisions of 3.6 and 4.3.2 or due to the fracture occuring under those conditions allowing for retests for elongation in Federal Test Method Standard No. 151, another specimen shall be machined from the same casting and tested. In the instance of failure due to an acceptable metal defect only one replacement specimen shall be permitted and failure of the replacement specimen to comply with the requirements of 3.4 shall be cause for rejection of the represented lot. Results of all tests, original and replacement tests, shall be recorded and specimens which failed to meet the requirements due to the above conditions shall be identified and retained by the contractor and made available to the procuring activity for examination upon request.

4.4.2 Soundness.-When specified by the procuring agency, castings will be inspected by ultrasonics, or if the geometry of the casting does not permit the use of ultrasonics, some form of radiography and Magnaglo, if approved by the procuring activity. Inspection Methods for less critical components shall be as specified by the procuring activity.

4.4.3 Mottle.-The samples selected in accordance with 4.3.3 shall be broken in such a manner that the fracture will extend through the heaviest section. The fracture shall be visually examined for compliance with 3.8.

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5. PREPARATION FOR SHIPMENT

5.1 Packing

5.1.1 Level C.-Castings shall be packed in accordance with the requirements of Specification MIL-STD-1169.

5.2 Labeling and Marking.-Labeling and marking shall be in accordance with the requirements of Specification MIL-STD-1169.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Soundness (see 3.6).

Custodian:
Army - MU

Preparing Activity:
Army - MU

Project Number: 1340-A-385

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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